Phase 2 - Information Gathering

3 Information Gathering

3.1 Data Sources

All the data necessary for this study are derived from multiple established bioinformatics and biomedical repositories. These sources collectively provide comprehensive information on oncogenic viruses, including Hepatitis B virus (HBV), Human herpesvirus 8 (HHV-8), Epstein–Barr virus, B95 strain (EBV), Human papillomavirus strains 16 and 18 (HPV16, HPV18), and Human T-cell lymphotropic virus strains 1A and 1C (HTLV-1A, HTLV-1C), the viral proteins they produce, and their interactions with human host proteins. The following databases were integrated to construct the final dataset:

3.1.1 NCBI Virus Database

This dataset provides gene annotations, coding sequences (CDS), and protein products, serving as a foundation for linking viral genes to their protein functions and host targets.

3.1.2 UniProt

Supplies detailed protein-level information, including molecular functions, biological processes, and subcellular localization. Both viral and human host proteins are retrieved from UniProt to ensure consistent identifier mapping and functional annotation.

3.1.3 IntAct Molecular Interaction Database

Contains experimentally validated molecular interactions, particularly virus–host protein–protein interactions (PPIs).

3.1.4 KEGG (Kyoto Encyclopedia of Genes and Genomes)

Provides pathway and functional mapping for host proteins. KEGG data enable the contextualization of interactions within cellular pathways, facilitating interpretation of how viral infections disrupt host signaling and metabolic processes involved in cancer development.

3.1.5 ViralZone (Swiss Institute of Bioinformatics)

This database provides biological context for each virus, helping to interpret how molecular interactions correspond to infection strategies and oncogenic potential.

3.1.6 World Health Organization (WHO)

Supply authoritative epidemiological data and official classifications of oncogenic viruses. WHO resources about recommended vaccines, allowing integration of public health information with molecular and interaction-level findings.

Each of these datasets contributes specific and complementary information:

- **Genomic and Protein Features (NCBI Virus, UniProt)**: gene, protein ids, and functional annotation.
- **Interaction Networks (IntAct)**: virus–host PPIs with evidence codes and literature references.
- **Pathway Integration (KEGG)**: mapping of viral effects on host cellular systems.
- **Biological Context (ViralZone)**: biological information about the viruses
- **Epidemiological and Clinical Data (WHO)**: classification and global impact of oncogenic viruses, recommended vaccines

Together, these resources enable the construction of an integrated dataset encompassing viral–host molecular interactions, functional annotations, and cancer associations, forming the foundation for downstream analyses and predictive modeling of virus-induced carcinogenesis.

3.2 Data Preprocessing

During the data preprocessing phase, interaction information was collected for seven viral entities, derived from five major oncogenic viruses: HBV, HHV-8, EBV (B95 strain), and multiple strains of HPV (HPV16 and HPV18) and HTLV (HTLV-1A and HTLV-1C). Virus–host PPI data were retrieved from the IntAct Molecular Interaction Database. All extraction, merging, and compilation steps were automated using a custom Python script to ensure efficiency, reproducibility, and comprehensive collection of interaction records. There was no misalignment in the data-gathering process. Following the generation of the fully merged IntAct dataset, an extensive data-cleaning procedure was applied to improve accuracy and ensure consistent annotation. In this step, 2,360 interaction records were removed due to missing required fields, and an additional 148 records were discarded as duplicates, defined as entries sharing the same combination of virus code, viral UniProt ID, and host UniProt ID. Only complete, unique, and biologically relevant interactions were retained for further analysis.

Following this cleanup, all remaining viral and host protein identifiers were cross-referenced with the UniProt database to retrieve detailed protein-level information, including functional annotations, sequence identifiers, and associated biological roles. Entries lacking sufficient UniProt information were excluded, resulting in the removal of 4 viral proteins and 82 host proteins. The final interaction dataset obtained after these filtering steps contained 31 distinct viral proteins and 759 human host proteins, representing a curated and well-validated set of interactions.

Subsequently, each host protein was mapped to its associated biological pathways using the KEGG and UniProt APIs in Python-based annotation workflows. The resulting pathway dataset underwent an additional round of cleaning to remove redundant, ambiguous, or incomplete pathway entries. Before this cleaning step, the pathway dataset consisted of 2,892 records; after filtering, 2,272 high-quality records remained, with 620 entries removed due to missing information or lack of clear annotation.

Since individual PPIs are not represented as entities in our knowledge graph, the ppi_data.tsv file, available in our GitHub repository, was used only as a source for generating protein-level information. In addition, because individual viruses can be associated with multiple cancers (and vice-versa), we included a dedicated lookup table, virus_cancer_lookup_table.tsv, to represent this relationship.

All final, cleaned, and biologically validated datasets have been saved in the repository as in the following table:

File	**Content**	**Related Etype**
cancer data.tsv	cancerID, cancerType, primarySite	Cancer
host_protein_data.tsv	hostUniprotID, hostGeneName, hostProteinName	HostProtein
viral_protein_data.tsv	viralUniprotID, viralGeneName, viralProteinName, virusCode	ViralProtein
protein_annotation_data.tsv	uniprotID, function, localization	ProteinAnnotation
vaccine_data.tsv	vaccineID, vaccineName, vaccineDescription, vaccineSource, targetVirus	Vaccine
virus_data.tsv	virusCode, organismName, species, host	Virus
virus_phenotype_data.tsv	phenotypeID, virusType, phenotypeSource, trait, virusName	Phenotype
pathway_data.tsv	pathwayKeggID, pathwayName, pathwayCategory, pathwayDescription, pathwaySource, hostUniprotID	Pathway

The GitHub repository contains several Python scripts used during the preprocessing and annotation of virus–host protein interactions:

1. **`intact_datasets_merger.py`**: Merging of virus–host PPI records from the IntAct Molecular Interaction Database; producing a single dataset.
2. **`ppi_information_extractor.py`**: Automates the extraction of PPI information from the fully merged intact dataset, and excluding any duplicated records or ones with missing data.
3. **`viral_and_host_protein_extractor.py`**: Extracts and compiles the final set of viral and host proteins for downstream analysis, ensuring that only complete, unique, and biologically relevant proteins are included.

4. `protein_annotation_extractor.py`: Retrieves detailed protein-level information for viral and host proteins by cross-referencing UniProt. It collects functional annotations, sequence identifiers, and associated biological roles.
5. `pathway_information_extractor.py`: Maps host proteins to their associated KEGG biological pathways via the KEGG and UniProt APIs.

3.3 Qualitative Evaluation

Throughout the information gathering phase, the ER model was revisited and refined to ensure coherence with the collected datasets. Several attributes that were no longer meaningful or relevant were removed, while others were renamed to improve clarity and consistency with biological terminology and data sources. Although certain relationship labels between ETypes were adjusted for precision, all previously defined relationships were retained, indicating that the original structural assumptions remained valid. Importantly, the informal purpose definition required no modification, demonstrating that the scope and objectives established in the purpose definition phase continued to align well with the data and domain understanding gained during this phase.